ALGORITHM

Kth Largest Element in a Binary Search Tree (BST)

Algorithm:

1. Start.

2. Read the number of elements, N, from the user.

3. Read N distinct integer values.

4. Create an empty Binary Search Tree (BST).

5. Insert each input value into the BST according to BST rules: smaller values go to the left subtree and larger values go to the right subtree.

6. Read the value K from the user.

7. Check whether K is valid. If K < 1 or K > N, display “Invalid value of K” and stop.

8. Initialize a counter to 0 and a variable to store the answer.

9. Perform reverse inorder traversal of the BST: visit the right subtree, then the root node, and then the left subtree.

10. Each time a node is visited, increase the counter by 1.

11. If the counter becomes equal to K, store the current node’s value as the Kth largest element.

12. Display the Kth largest element.

13. Stop.

Traversal order: Right → Root → Left (visits BST values in descending order).